

Brainstorming

What makes a good opportunity for Praxis I?

- Specific -> narrow, converge
- Diverse ways to solve
- Feasibility for first year students
 - Costs
 - Skills
 - etc.
- Interesting to all the internal stakeholders involved (EngSci's for this project)
- Real problem that hasn't been solved (google to double check)
 - Look at reference designs or handbooks etc.

Potential Opportunities:

- STANGEBY'S CHALK
- Uncomfortable chairs
 - Chairs are creaky, wobbly
 - SO hard to get in and sit with bumping the person next to you
- Tables are hard to stow/unstow without hitting the person next to you
- Professors don't talk loud enough for people in the back to hear
 - Pocket mics for lectures
- Midterm schedules and assignment due dates aren't unified (they're spread out on 6 different syllabi)
 - Quercus google extension that takes the data from all the syllabi and combines into a centralized exam/assignment calendar
- Slow elevators in Myhal
 - THEY GO DOWN TO PARKING THEN BACK UP
- Water fountains
 - Taste odd
 - Rarely changed when needed
 - Not working half the time
 - Not enough water bottle refill stations
 - Water pressure isn't high enough

Homework: each individual chooses one DfX and identifies 4 more opportunities that can be defined using their DfX. Bring these opportunities to next studio

Useability

- We have to take the TCard out of our bags and/or the plastic case for the TCard each time we need to swipe it
 - Very easy thing to fix. Can be done by buying something

- Binder explosion -> nvm
 - Not everyone had this problem
- Wet umbrellas and wet shoes during winter, making the classroom all wet, seats & tables etc
 - Buy stuff to fix it
- Headphones pinching
 -

Environment

- Printers print out a title page every time for printing
 - Software fix
- Some textbooks are still physical
 - Not very good splartz
- Paper towels offered in bathrooms
 - Lots of people want paper towels
- Food trucks give one-time-use utensils
 - Maybe some affiliation with Uoft and food trucks
- A lot of food waste at dining halls
 - Maybe?

Repairability

- Unrepairable personal electronics
 - Apple is very notorious for being against repairability with their very shut down and soldered design for everything on their macs, forcing students to pay very high fees to get Apple to fix it or pay for a new one. There is a very large percentage of engineering science students that use Apple electronics and almost all engineering science students use Apple iPads
 - These devices are repairable to an extent given the expertise and tools
 - What can be done the student end is that they are educated on how to repair these very complex electronics for easy fixes
- Wear and tear on personal transportation equipment
- Dorm Appliances -> breaking down
- Everyday small tech stuff
 - Cables, etc.

Safety

- Better traffic arrangement
-

Homework

- Repairability
- Maintainability
- Testing
- Assembling

- The environment
- Durability
- Reliability
- Manufacturing
- Recycling
- Logistics
- Usability
- Affordability

New Opportunity Ideas

- Improving biking opportunities? - Greennn
 - Increasing biking slots/return area
- Jay walking on Campus -> St George street
 - Bridge
 - Tunnel
 - More traffic lights
 - Close the road
- Umbrellas flipping inside out during big winds

Parts of Life

- Eating
 - Dining halls, restaurants, food trucks, meal prep
- Cleaning/Hygiene
 - Vacuums
 - Hair on the floor
 -
- Sleeping
 - Alarms that we forget to set? (probably software though)
 - **Roommates in dorms sleeping at different times difficulty sleeping?**
 - **Need: a way for roommates to not disturb each other when the other is sleeping (might be too broad)**
 - Noise cancelling bed
 - Pullout slideable soundproof wall (sound dampening foam)
 - Directional alarm maybe in the wall
 - Vibrating bed to wake you up
- Travelling
 - Public transport, green transport options
 - Traffic
 - Weather
 -
- Studying
- Academic
 - Lectures

- Quizzes, exams, homework
- Pens not writing
- Social
 - Clubs
 - Design teams
- Entertainment

Workplan:

Research

Writing

Further Observation

Select another opportunity

Opportunity	Solutions
Binder Inconveniences - Need a good way to store papers? - Binder rings opening unwanted when attempting to flip large amounts of paper at once - Binder hole misalignment - Its annoying to hole punch	- Change the opening/closing mechanism - Change the binder rings so they themselves can hole punch -
Broken electronics are wasted	- Make a course to educate people about how to repair their own devices. Make it a standard not just something that is available online - Have supplies to repair devices on campus - Reuse broken electronics for educational purposes -
Food trucks give one-time-use utensils - Generates a lot of waste from every customer - Need a better way for people to eat without using one-time utensils	- Reusable tray system + designated eating area for food truck food - Give food trucks recyclable containers and financial incentive to use them - Everyone that brings their own utensils gets a discount (funded by UofT)
Resting between classes?	

Opportunities

A good opportunity:

i. reflects the team's and primary stakeholders' interests and values,

- ii. represents a stakeholder need (as demonstrated by observation and research),
- iii. is solvable by a team of first-year engineering science students,
- iv. can be represented and solved in multiple meaningful ways (e.g. calculation, physical prototype(s), drawings, demonstrations, or other means), but
- v. cannot simply be solved by selecting an existing design

Oct 4 Meeting Goals:

- Put a timer for everything
- 20 min brainstorm, broken into 5 min intervals
- 15 min to research the three ideas narrowed down to
 - Look specifically at reference designs
- Discuss for 10 min to finalize a single idea
 - Keep it simple
 - Focus on the biggest problem related to that one idea
- Plan out the project for the next week in the remaining time

To Do List

Finish By Saturday

- Each person should bring 4 opportunities or one really good opportunity
 - Each with 5 solutions
- Read the design brief doc